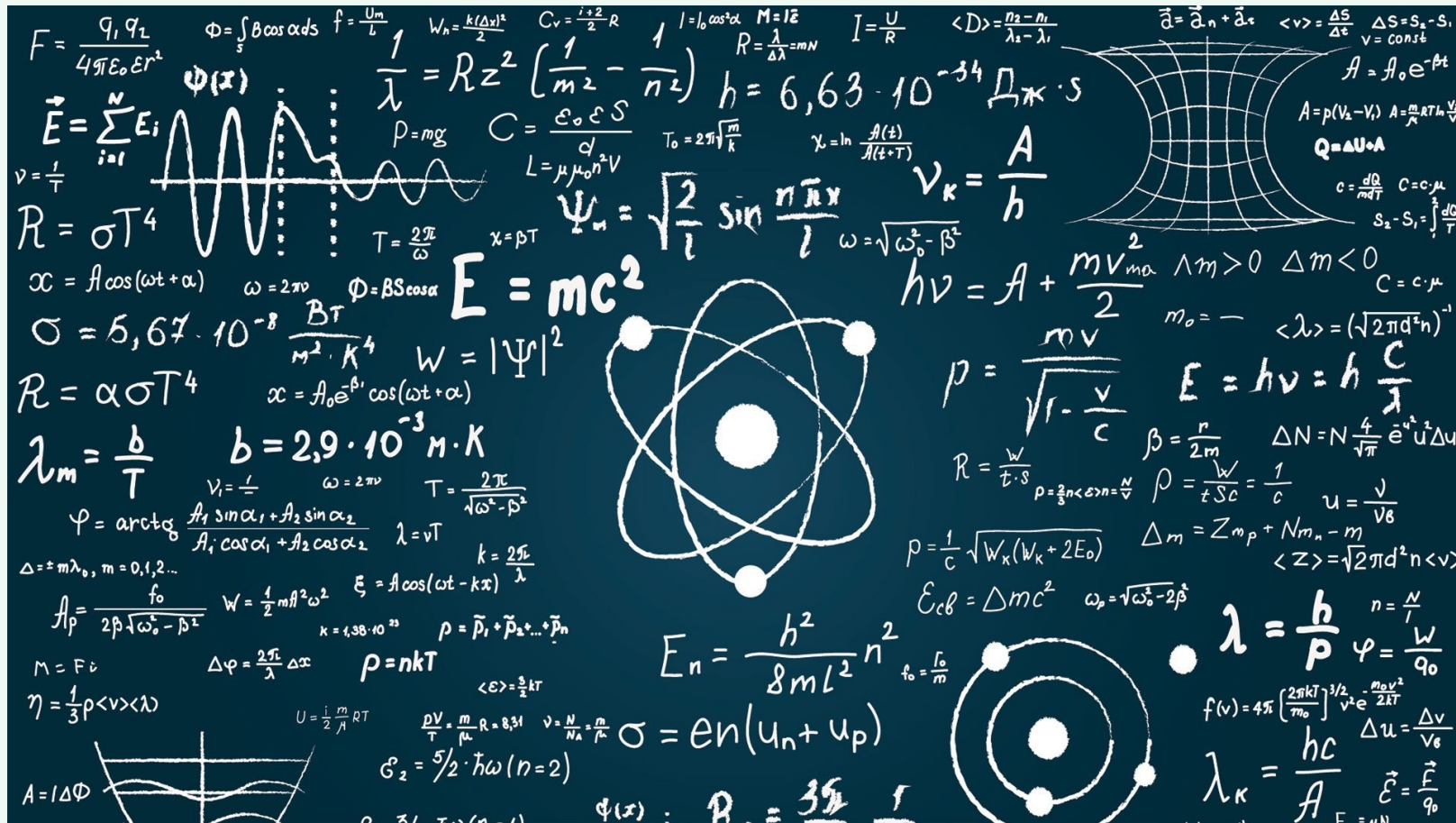


Quantum Mechanics 2



Lecture 4
(Autumn Term, 2025)

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Recap of Lecture 3

QM postulate 1: wavefunction is a complete description of the state of the quantum system, containing all the physical information it is possible to know about the object.

QM Postulate 2: For every physical observable, there is an associated mathematical operator.

We defined Probability Density Function $P(x, t) = |\psi|^2 = \psi^* \psi$

and Expectation Value $\langle f(x) \rangle = \int_{-\infty}^{\infty} \psi^* f(x) \psi dx$

QM operators extract physical information from wavefunction

Momentum $\hat{p} = -i\hbar \frac{\partial}{\partial x}$

Kinetic Energy $\frac{\hat{p}^2}{2m} = \frac{-\hbar^2}{2m} \frac{\partial^2}{\partial x^2}$

Total energy $\hat{E} = i\hbar \frac{\partial}{\partial t}$

QM operators extract physical information from wavefunction:

$$\psi = Ae^{i(kx - \omega t)} \quad \Rightarrow \quad \hat{p}\psi = \hbar k\psi$$

This is an example of **eigenvalue equation**

$$\hat{O}\psi_i = a_i\psi_i \quad \text{operator (function) = constant x function}$$

ψ_i is an eigenfunction of $\hat{O}$, describing quantum state (eigenstate) with definite value of O , the observable corresponding to operator $\hat{O}$

a_i is an eigenvalue of $\hat{O}$, the value of O that would be measured for eigenstate ψ_i

Complete set of all eigenvalues gives all possible results of measurements of observable O , for any quantum state

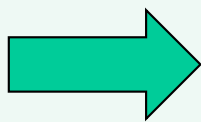
Expectation value for any observable O , corresponding to operator $\hat{O}$, for wavefunction ψ gives mean result of **many measurements**:

$$\langle \hat{O} \rangle = \int_{-\infty}^{\infty} \psi^* \hat{O} \psi dx$$

$$\text{S.E.} \quad \frac{-\hbar^2}{2m} \frac{\partial^2 \psi(x,t)}{\partial x^2} + V(x,t)\psi(x,t) = i\hbar \frac{\partial \psi(x,t)}{\partial t}$$

In many important cases, potential energy function is constant in time: $V(x,t) = V(x)$

Can simplify S.E. using method of separation of variables on partial differential equation.



Time-Independent
Schrodinger's equation
(T.I.S.E.)

$$\left[\text{Note: order matters} \quad \psi^* \frac{\partial \psi}{\partial x} \neq \frac{\partial \psi^*}{\partial x} \psi \neq \frac{\partial \psi^* \psi}{\partial x} \right]$$

Summary

Eigenvalue equation: $\hat{O}\psi_i = a_i\psi_i$

ψ_i is an eigenfunction of $\hat{O}$, describing quantum state (eigenstate) with definite value of O , the observable corresponding to operator $\hat{O}$

a_i is an eigenvalue of $\hat{O}$, the value of O that would be measured for eigenstate ψ_i

Expectation value as result of many measurements:

$$\langle \hat{O} \rangle = \int_{-\infty}^{\infty} \psi^* \hat{O} \psi dx$$

For $V=\text{constant}$, obtained T.I.S.E.

Free-particle solution $\Psi = [Ae^{ikx} + Be^{-ikx}]e^{-iEt/\hbar}$

$$k = \frac{\sqrt{2m(E - V_0)}}{\hbar}$$

Next week: explore the case of a potential step